

Name of the Program: B.Tech.
Semester: I
Name of the Course: Basic Electrical Engineering
Course Code: TEE-101
Time: 1.5 hours

Maximum Marks: 25

Note:

- (i) This QP has 3 sections. Attempt questions from each Section.
- (ii) Answer the questions as per the instructions in each section.

Section A

This section has MCQ's. Attempt any 5. Each question carries 1 mark.

1. The resistance of a conductor depends on:

(1 Mark) CO1

- A) Length
- B) Cross-sectional area
- C) Material
- D) All of the above

2. Three resistors of $6\ \Omega$ each are connected in parallel. Their equivalent resistance is: (1 Mark) CO1

- A) $18\ \Omega$
- B) $6\ \Omega$
- C) $3\ \Omega$
- D) $2\ \Omega$

3. Kirchhoff's Current Law (KCL) is based on conservation of:

(1 Mark) CO1

- A) Energy
- B) Charge
- C) Momentum
- D) Power

4. The internal resistance of an ideal voltage source is:

(1 Mark) CO1

- A) Infinite
- B) $1\ \Omega$
- C) Zero
- D) $100\ \Omega$

5. The principle of superposition can be applied to:

(1 Mark) CO2

- A) Linear circuits
- B) Non-linear circuits only
- C) Mechanical circuits
- D) Magnetic materials only

6. The relation between Thevenin and Norton resistance is:

(1 Mark) CO2

- A) $R_{th} = 1/R_N$
- B) $R_{th} = R_N$
- C) $R_{th} = R_N^2$
- D) $R_{th} = 0$



Term Evaluation (Odd) Semester Regular Examination September 2026

(1 Mark) CO2

7. At maximum power transfer, the efficiency of a DC circuit is:

- A) 25%
- B) 50%
- C) 75%
- D) 100%

8. A $10\ \Omega$ resistor is connected across a 20 V supply. If the resistance is doubled while the voltage remains constant, the current becomes:

(1 Mark) CO3

- A) Doubled
- B) Halved
- C) Four times
- D) Unchanged

Section B

Attempt any 2 questions out of given 4 questions. Each question carries 5 marks.

Q1. Consider a 230 V, 100 W incandescent lamp. Determine:

(5 Marks) CO3

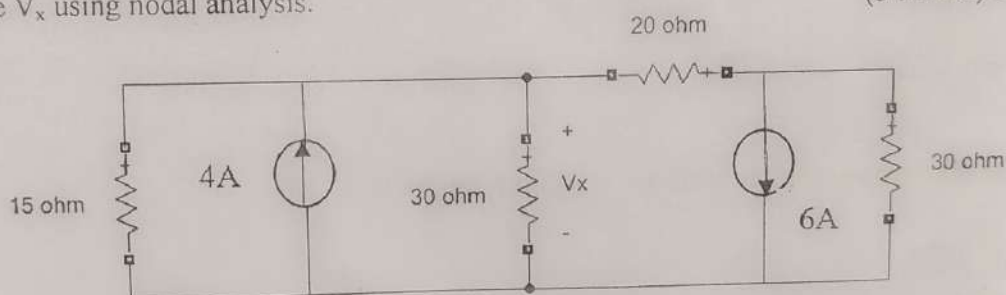
- a) the lamp resistance
- b) the lamp current
- c) the energy consumed by the lamp in 8 hours.

Q2. In an inductor L Henry, charge is flowing at the rate of q coulomb/Sec. What is the stored energy?

(5 Marks) CO2

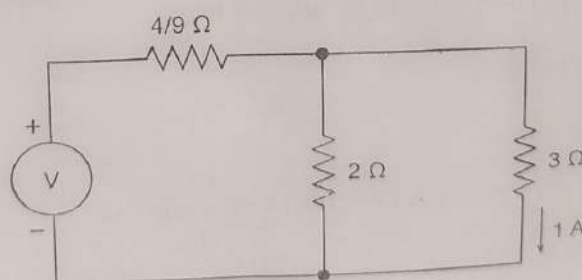
Q3. Determine V_x using nodal analysis.

(5 Marks) CO3



Q4. For the circuit, find the Voltage V.

(5 Marks) CO3

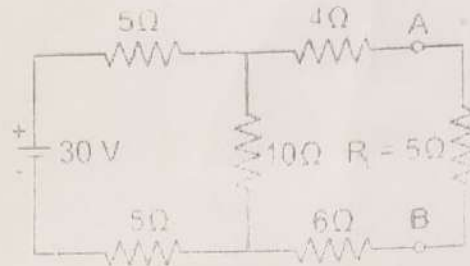




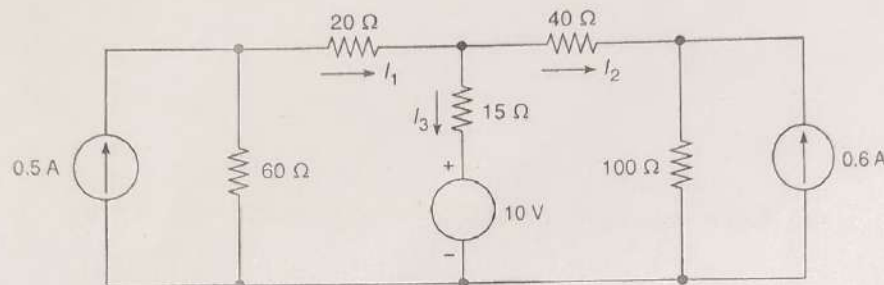
Term Evaluation (Odd) Semester Regular Examination September 2026
Section C

Attempt any 2 questions out of given 4 questions. Each question carries 5 marks.

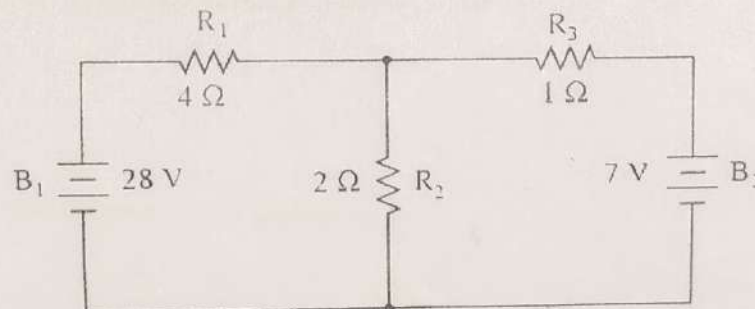
Q1 In the network shown in figure, calculate the current through the load resistor R_L by using Norton's Theorem.
(5 Marks) CO3



Q2. Using mesh analysis find currents I_1 , I_2 and I_3 in the circuit. Also find the power supplied by the two current sources.
(5 Marks) CO3



Q3. Find the current through 2 ohm using the principle of Superposition
(5 Marks) CO3



Q4. What are the different types of electrical sources? Explain ideal and practical voltage and current sources with suitable circuit diagrams.
(5 Marks) CO2